Proof by Mathematical Induction

ANSWER KEY

The three steps of induction

1 State the three steps required for a valid proof by mathematical induction.

(1) Base case: verify the statement holds for the smallest value of n (usually $n = 1$). (2) Inductive hypothesis: assume the statement holds for $n = k$. (3) Inductive step: use the assumption to prove the statement holds for $n = k + 1$. Concluding, by the principle of mathematical induction, the statement holds for all n in the given range.

Proving a summation formula

2 Prove by induction that $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$ for all $n \geq 1$.

Base case ($n = 1$): LHS $= 1$. RHS $= \frac{1(2)}{2} = 1$. True. Inductive hypothesis: assume $\sum_{r=1}^{k} r = \frac{k(k+1)}{2}$. Inductive step: $\sum_{r=1}^{k+1} r = \frac{k(k+1)}{2} + (k+1) = (k+1)\left(\frac{k}{2} + 1\right) = \frac{(k+1)(k+2)}{2}$, which matches the formula with $n = k + 1$. Since it holds for $n = 1$ and $n = k \Rightarrow n = k + 1$, by induction it holds for all $n \geq 1$.

3 Prove by induction that $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \geq 1$.

Base case ($n = 1$): LHS $= 1$. RHS $= \frac{1(2)(3)}{6} = 1$. True. Assume true for $n = k$: $\sum_{r=1}^{k} r^2 = \frac{k(k+1)(2k+1)}{6}$. For $n = k + 1$: $\sum_{r=1}^{k+1} r^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2 = (k+1)\left[\frac{k(2k+1)}{6} + (k+1)\right] = (k+1) \cdot \frac{2k^2 + 7k + 6}{6} = \frac{(k+1)(k+2)(2k+}{6}$, matching the formula at $n = k + 1$. By induction, true for all $n \geq 1$.

Proving divisibility results

4 Prove by induction that $7^n - 1$ is divisible by 6 for all $n \geq 1$.

Base case ($n = 1$): $7^1 - 1 = 6$, divisible by 6. Assume $7^k - 1 = 6m$ for some integer m, so $7^k = 6m + 1$. For $n = k + 1$: $7^{k+1} - 1 = 7 \cdot 7^k - 1 = 7(6m + 1) - 1 = 42m + 6 = 6(7m + 1)$, which is divisible by 6. By induction, $7^n - 1$ is divisible by 6 for all $n \geq 1$.

Induction with inequalities

5 Prove by induction that $2^n > n^2$ for all integers $n \geq 5$.

Base case ($n = 5$): $2^5 = 32$, $5^2 = 25$, and $32 > 25$. True. Assume $2^k > k^2$ for some $k \geq 5$. For $n = k + 1$: $2^{k+1} = 2 \cdot 2^k > 2k^2$. We need $2k^2 \geq (k+1)^2 = k^2 + 2k + 1$, i.e. $k^2 - 2k - 1 \geq 0$, which holds for $k \geq 3$ (true since $k \geq 5$). So $2^{k+1} > 2k^2 \geq (k+1)^2$. By induction, $2^n > n^2$ for all $n \geq 5$.

Induction in calculus: derivatives of a product

6 Prove by induction that $\frac{d^n}{dx^n}(e^{ax}) = a^n e^{ax}$ for all positive integers n.

Base case ($n = 1$): $\frac{d}{dx}(e^{ax}) = ae^{ax} = a^1 e^{ax}$. True. Assume $\frac{d^k}{dx^k}(e^{ax}) = a^k e^{ax}$. Then $\frac{d^{k+1}}{dx^{k+1}}(e^{ax}) = \frac{d}{dx}(a^k e^{ax}) = a^k \cdot ae^{ax} = a^{k+1}e^{ax}$, matching the formula at $n = k + 1$. By induction, true for all positive integers n.

Induction with a recursively defined sequence

7 A sequence is defined by $u_1 = 3$ and $u_{n+1} = 2u_n - 1$. Prove by induction that $u_n = 2^n + 1$ for all $n \geq 1$.

Base case ($n = 1$): $u_1 = 3$ (given). Formula: $2^1 + 1 = 3$. True. Assume $u_k = 2^k + 1$. Then $u_{k+1} = 2u_k - 1 = 2(2^k + 1) - 1 = 2^{k+1} + 2 - 1 = 2^{k+1} + 1$, matching the formula at $n = k + 1$. By induction, $u_n = 2^n + 1$ for all $n \geq 1$.

Proving a geometric-series-style summation

8 Prove by induction that the sum $3^0 + 3^1 + \cdots + 3^{n-1}$ equals $\frac{3^n - 1}{2}$ for all $n \geq 1$.

Base case ($n = 1$): LHS $= 3^0 = 1$. RHS $= \frac{3^1 - 1}{2} = \frac{2}{2} = 1$. True. Assume the sum $3^0 + \cdots + 3^{k-1} = \frac{3^k - 1}{2}$. For $n = k + 1$: adding the next term 3^k gives $\frac{3^k - 1}{2} + 3^k = \frac{3^k - 1 + 2 \cdot 3^k}{2} = \frac{3 \cdot 3^k - 1}{2} = \frac{3^{k+1} - 1}{2}$, matching the formula at $n = k + 1$. By induction, true for all $n \geq 1$.

9 Prove by induction that the sum $1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n!$ equals $(n + 1)! - 1$ for all $n \geq 1$.

Base case ($n = 1$): LHS $= 1 \cdot 1! = 1$. RHS $= 2! - 1 = 2 - 1 = 1$. True. Assume $1 \cdot 1! + \cdots + k \cdot k! = (k + 1)! - 1$. For $n = k + 1$: adding the next term $(k + 1)(k + 1)!$ gives $(k + 1)! - 1 + (k + 1)(k + 1)! = (k + 1)!\,[1 + (k + 1)] - 1 = (k + 1)!\,(k + 2) - 1 = (k + 2)! - 1$, matching the formula at $n = k + 1$. By induction, true for all $n \geq 1$.

Proving an inequality involving factorials

10 Prove by induction that $n! > 2^n$ for all integers $n \geq 4$.

Base case ($n = 4$): $4! = 24$, $2^4 = 16$, and $24 > 16$. True. Assume $k! > 2^k$ for some $k \geq 4$. For $n = k + 1$: $(k + 1)! = (k + 1) \cdot k! > (k + 1) \cdot 2^k$. Since $k \geq 4$, $k + 1 \geq 5 > 2$, so $(k + 1) \cdot 2^k > 2 \cdot 2^k = 2^{k+1}$. Combining, $(k + 1)! > 2^{k+1}$, matching the inequality at $n = k + 1$. By induction, $n! > 2^n$ for all $n \geq 4$.

Proving a divisibility result with two variables

11 Prove by induction that $n^3 + 2n$ is divisible by 3 for all positive integers $n \ge 1$.

Base case ($n = 1$): $1^3 + 2(1) = 3$, divisible by 3. Assume $k^3 + 2k = 3m$ for some integer m. For $n = k + 1$:

$(k+1)^3 + 2(k+1) = k^3 + 3k^2 + 3k + 1 + 2k + 2 = (k^3 + 2k) + 3k^2 + 3k + 3 = 3m + 3(k^2 + k + 1) = 3(m + k^2 +$

, which is divisible by 3. By induction, $n^3 + 2n$ is divisible by 3 for all $n \ge 1$.

12 Prove by induction that $9^n - 1$ is divisible by 8 for all positive integers $n \ge 1$, and explain briefly why a similar argument would show $a^n - 1$ is divisible by $a - 1$ for any integer $a > 1$.

Base case ($n = 1$): $9^1 - 1 = 8$, divisible by 8. Assume $9^k - 1 = 8m$ for some integer m, so $9^k = 8m + 1$. For $n = k + 1$: $9^{k+1} - 1 = 9 \cdot 9^k - 1 = 9(8m + 1) - 1 = 72m + 9 - 1 = 72m + 8 = 8(9m + 1)$, divisible by 8. By induction, true for all $n \ge 1$. The same inductive pattern works for any base a: $a^{k+1} - 1 = a \cdot a^k - 1 = a[(a-1)m' + 1] - 1 = (a-1)(am') + (a-1) = (a-1)(am' + 1)$, always a multiple of $a - 1$.

Induction with a recurrence relation for a sequence of sums

13 A sequence satisfies $S_1 = 2$ and $S_{n+1} = S_n + 6n + 2$ for $n \ge 1$. Prove by induction that $S_n = 3n^2 - n$ for all $n \ge 1$.

Base case ($n = 1$): $S_1 = 2$ (given). Formula: $3(1)^2 - 1 = 2$. True. Assume $S_k = 3k^2 - k$. Then $S_{k+1} = S_k + 6k + 2 = 3k^2 - k + 6k + 2 = 3k^2 + 5k + 2$. The formula at $n = k + 1$ gives $3(k+1)^2 - (k+1) = 3k^2 + 6k + 3 - k - 1 = 3k^2 + 5k + 2$, which matches exactly. Since it holds for $n = 1$ and $n = k \Rightarrow n = k + 1$, by induction $S_n = 3n^2 - n$ for all $n \ge 1$.

Induction with a strengthened hypothesis

14 Prove by induction that $11^n - 4^n$ is divisible by 7 for all positive integers $n \ge 1$.

Base case ($n = 1$): $11^1 - 4^1 = 7$, divisible by 7. Assume $11^k - 4^k = 7m$ for some integer m, so $11^k = 7m + 4^k$. For $n = k + 1$:

$11^{k+1} - 4^{k+1} = 11 \cdot 11^k - 4 \cdot 4^k = 11(7m + 4^k) - 4 \cdot 4^k = 77m + 11 \cdot 4^k - 4 \cdot 4^k = 77m + 7 \cdot 4^k = 7(11m + 4^k)$

, divisible by 7. By induction, $11^n - 4^n$ is divisible by 7 for all $n \ge 1$.

Induction to prove a formula for the sum of cubes

15 Prove by induction that $\sum_{r=1}^{n} r^3 = \left[\frac{n(n+1)}{2}\right]^2$ for all $n \ge 1$.

Base case ($n = 1$): LHS $= 1^3 = 1$. RHS $= \left[\frac{1(2)}{2}\right]^2 = 1^2 = 1$. True. Assume $\sum_{r=1}^{k} r^3 = \left[\frac{k(k+1)}{2}\right]^2$. For $n = k + 1$:

$\sum_{r=1}^{k+1} r^3 = \left[\frac{k(k+1)}{2}\right]^2 + (k+1)^3 = (k+1)^2\left[\frac{k^2}{4} + (k+1)\right] = (k+1)^2 \cdot \frac{k^2 + 4k + 4}{4} = (k+1)^2 \cdot \frac{(k+2)^2}{4} = \left[\frac{(k+1)(k}{2}\right.$

, matching the formula at $n = k + 1$. By induction, true for all $n \ge 1$.

16 Prove by induction that $5^n + 2 \cdot 3^{n-1} + 1$ is divisible by 8 for all positive integers $n \geq 1$.

Base case ($n = 1$): $5^1 + 2(3^0) + 1 = 5 + 2 + 1 = 8$, divisible by 8. Assume $5^k + 2 \cdot 3^{k-1} + 1 = 8m$ for some integer m. Consider the difference between consecutive terms: $[5^{k+1} + 2 \cdot 3^k + 1] - [5^k + 2 \cdot 3^{k-1} + 1] = 5^{k+1} - 5^k + 2 \cdot 3^k - 2 \cdot 3^{k-1} = 4 \cdot 5^k + 4 \cdot 3^{k-1} = 4(5^k + 3^{k-1})$. Since 5^k and 3^{k-1} are both odd, their sum is even, so $4(5^k + 3^{k-1})$ is a multiple of 8. Therefore $5^{k+1} + 2 \cdot 3^k + 1 = [5^k + 2 \cdot 3^{k-1} + 1] + 4(5^k + 3^{k-1}) = 8m + (\text{a multiple of } 8)$, which is also a multiple of 8. By induction, the statement holds for all $n \geq 1$.

Induction to prove a formula for a weighted geometric sum

17 Prove by induction that $\sum_{r=1}^{n} r \cdot 2^r = (n-1)2^{n+1} + 2$ for all $n \geq 1$.

Base case ($n = 1$): LHS $= 1 \cdot 2^1 = 2$. RHS $= (1-1)2^2 + 2 = 0 + 2 = 2$. True. Assume $\sum_{r=1}^{k} r \cdot 2^r = (k-1)2^{k+1} + 2$. For $n = k + 1$: adding the next term $(k+1)2^{k+1}$ gives $(k-1)2^{k+1} + 2 + (k+1)2^{k+1} = [(k-1) + (k+1)]2^{k+1} + 2 = 2k \cdot 2^{k+1} + 2 = k \cdot 2^{k+2} + 2$. The formula at $n = k + 1$ requires $[(k+1) - 1]2^{k+2} + 2 = k \cdot 2^{k+2} + 2$, which matches exactly. By induction, the formula holds for all $n \geq 1$.

A closing check: identifying an invalid induction argument

18 A student claims to prove that all horses are the same color using induction on the number of horses n in a group: the base case $n = 1$ is trivially true (one horse is trivially "the same color" as itself), and the student then argues that if any k horses are the same color, then any $k + 1$ horses must also be the same color. Explain precisely where this argument breaks down, and state why a correct induction proof must avoid this flaw.

The flaw is in the inductive step: the argument tries to overlap two groups of k horses (the first k and the last k out of $k + 1$ total) and conclude they share a common color through the overlap. This overlap argument only works when the overlap is non-empty, which requires $k \geq 2$. It fails silently at the transition from $n = 1$ to $n = 2$, since two groups of 1 horse each (drawn from a total of 2) have no horse in common, so there is no shared reference point linking their colors. This illustrates why a correct induction proof must verify that the inductive step logic genuinely holds for the smallest values of k used in the argument, not just assume it holds uniformly for all k.